



UNIVERSITY OF ENGINEERING AND TECHNOLOGY MARDAN

Department: Computer Sceince

Course: Mobile App Develepment

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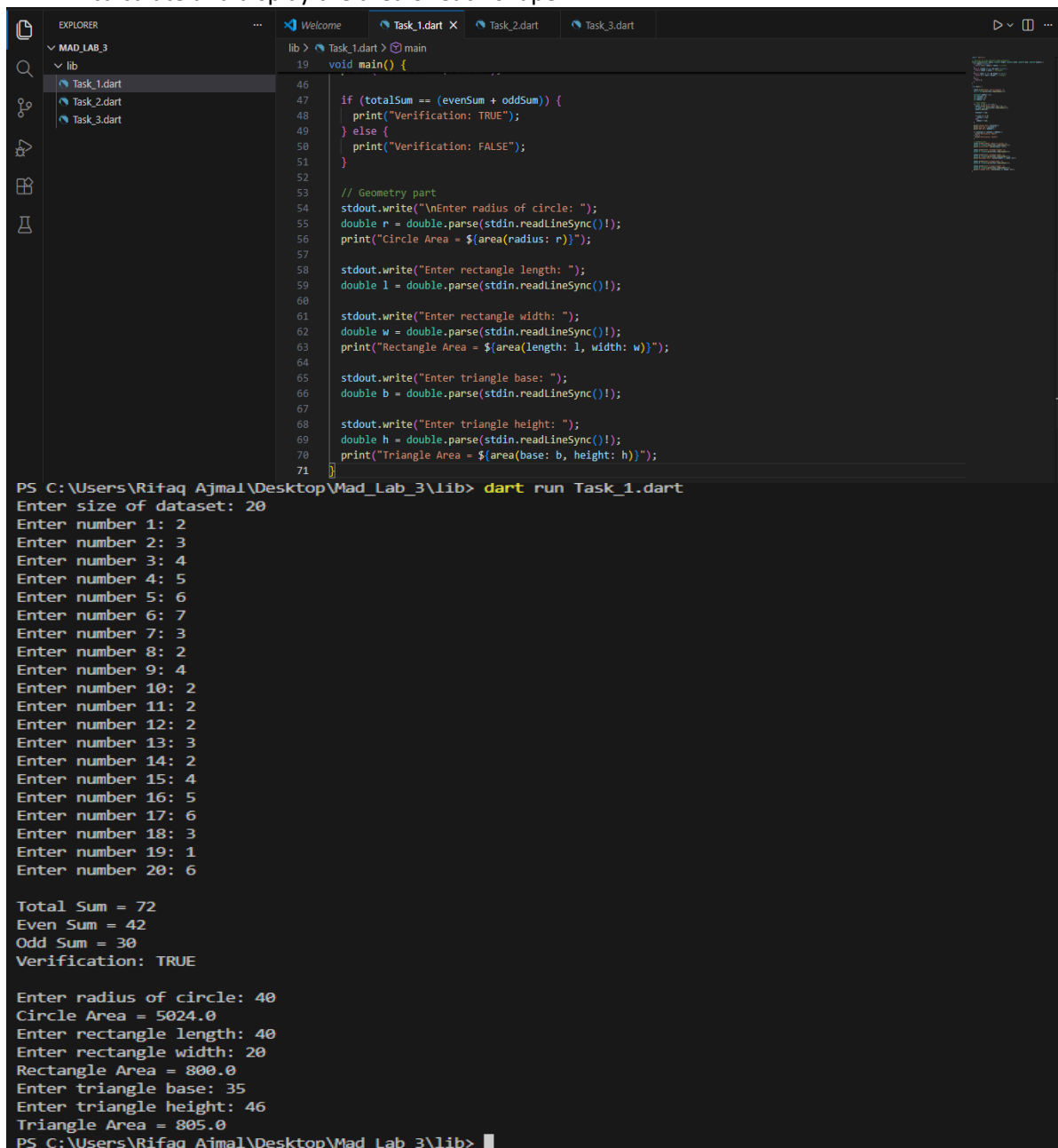
Stream: Computer Science (Section A)

Date : 3 Feb 2026

Lab : 3

Task 1:

- 1) Dart program for a "Geometry & Data Analysis" tool. The program should first ask the user to provide a positive integer representing the size of a dataset. Based on this size, the program must collect a list of numbers from the user, calculate the total sum of the list, and find the separate sums of even and odd numbers within that list. After displaying these results, the program should verify and print whether the total sum is mathematically equal to the combined sum of even and odd numbers. Finally, to integrate geometric calculations, the program must prompt the user for the dimensions of a circle (radius), a rectangle (length and width), and a triangle (base and height), then use a single function with optional named parameters to calculate and display the area of each shape.



The screenshot shows an IDE with a file explorer on the left containing 'MAD_LAB_3' and 'lib' folders. The main editor displays 'Task_1.dart' with the following code:

```
lib > Task_1.dart > main
19 void main() {
20
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47 if (totalSum == (evenSum + oddSum)) {
48   print("Verification: TRUE");
49 } else {
50   print("Verification: FALSE");
51 }
52
53 // Geometry part
54 stdout.write("\nEnter radius of circle: ");
55 double r = double.parse(stdin.readLineSync());
56 print("Circle Area = ${area(radius: r)}");
57
58 stdout.write("Enter rectangle length: ");
59 double l = double.parse(stdin.readLineSync());
60
61 stdout.write("Enter rectangle width: ");
62 double w = double.parse(stdin.readLineSync());
63 print("Rectangle Area = ${area(length: l, width: w)}");
64
65 stdout.write("Enter triangle base: ");
66 double b = double.parse(stdin.readLineSync());
67
68 stdout.write("Enter triangle height: ");
69 double h = double.parse(stdin.readLineSync());
70 print("Triangle Area = ${area(base: b, height: h)}");
71 }
```

The terminal output below the code shows the program's execution:

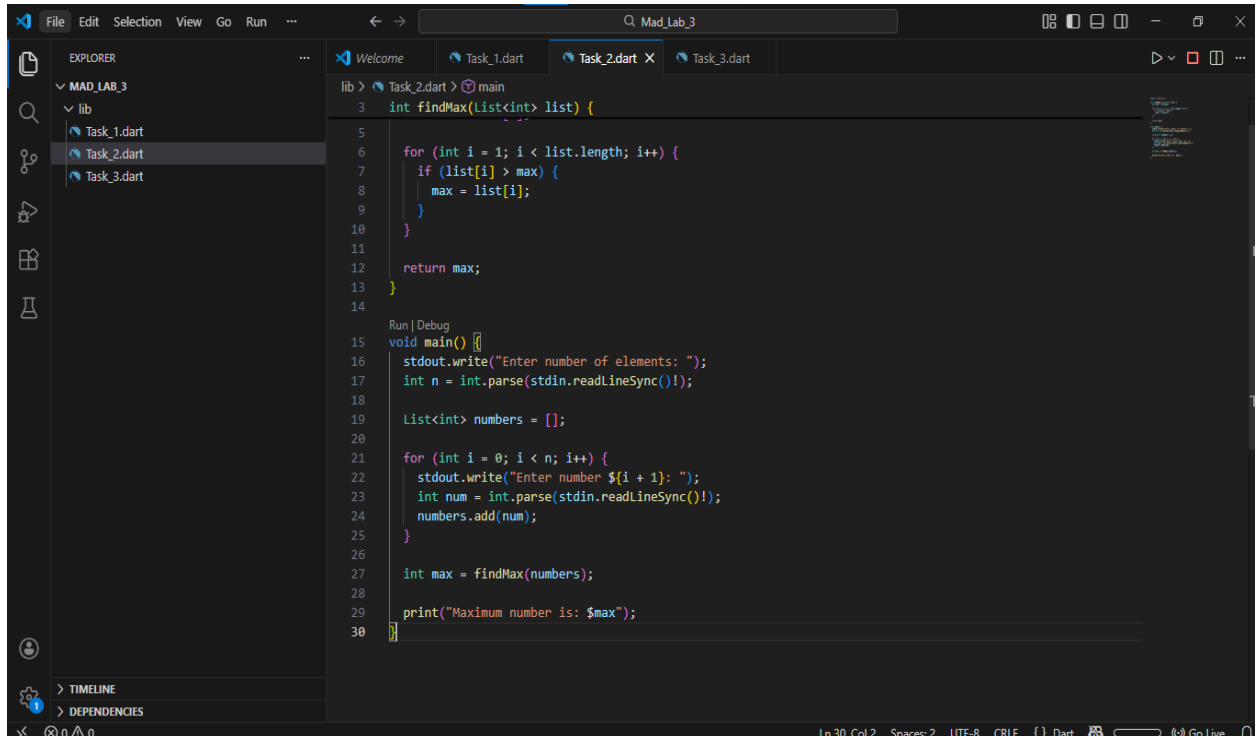
```
PS C:\Users\Rifaq Ajmal\Desktop\Mad_Lab_3\lib> dart run Task_1.dart
Enter size of dataset: 20
Enter number 1: 2
Enter number 2: 3
Enter number 3: 4
Enter number 4: 5
Enter number 5: 6
Enter number 6: 7
Enter number 7: 3
Enter number 8: 2
Enter number 9: 4
Enter number 10: 2
Enter number 11: 2
Enter number 12: 2
Enter number 13: 3
Enter number 14: 2
Enter number 15: 4
Enter number 16: 5
Enter number 17: 6
Enter number 18: 3
Enter number 19: 1
Enter number 20: 6

Total Sum = 72
Even Sum = 42
Odd Sum = 30
Verification: TRUE

Enter radius of circle: 40
Circle Area = 5024.0
Enter rectangle length: 40
Enter rectangle width: 20
Rectangle Area = 800.0
Enter triangle base: 35
Enter triangle height: 46
Triangle Area = 805.0
PS C:\Users\Rifaq Ajmal\Desktop\Mad_Lab_3\lib>
```

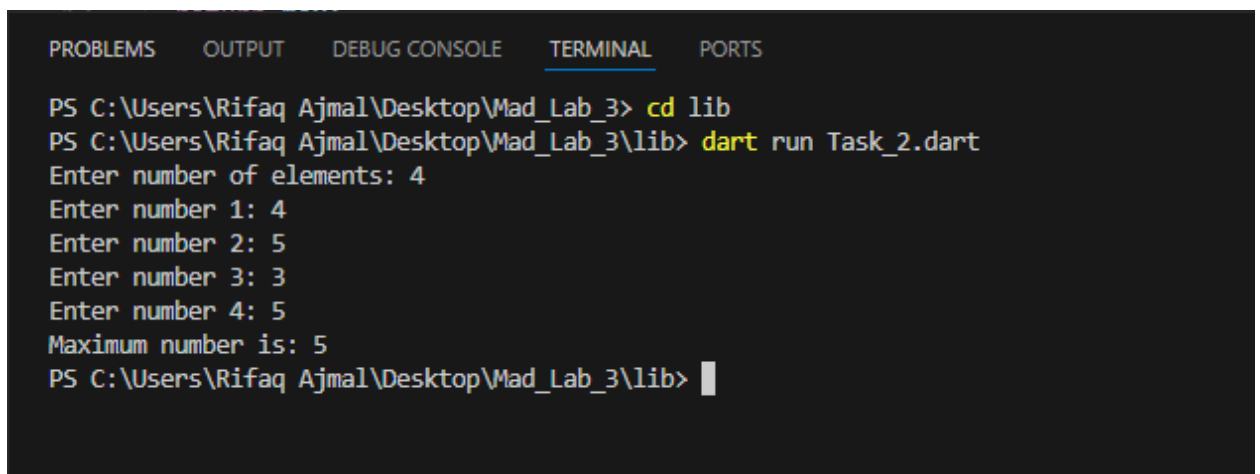
Task 2

2) Using function find the max no form list?



The screenshot shows an IDE with a project named 'MAD_LAB_3'. The file explorer on the left shows a 'lib' directory containing 'Task_1.dart', 'Task_2.dart', and 'Task_3.dart'. The main editor displays the code for 'Task_2.dart'. The code defines a function 'findMax' that takes a list of integers and returns the maximum value. It also includes a 'main' function that prompts the user to enter the number of elements, reads the input, creates a list, prompts for each element, and prints the maximum value found.

```
lib > Task_2.dart > main
3  int findMax(List<int> list) {
4
5
6      for (int i = 1; i < list.length; i++) {
7          if (list[i] > max) {
8              max = list[i];
9          }
10     }
11
12     return max;
13 }
14
Run | Debug
15 void main() {
16     stdout.write("Enter number of elements: ");
17     int n = int.parse(stdin.readLineSync()!);
18
19     List<int> numbers = [];
20
21     for (int i = 0; i < n; i++) {
22         stdout.write("Enter number ${i + 1}: ");
23         int num = int.parse(stdin.readLineSync()!);
24         numbers.add(num);
25     }
26
27     int max = findMax(numbers);
28
29     print("Maximum number is: $max");
30 }
```



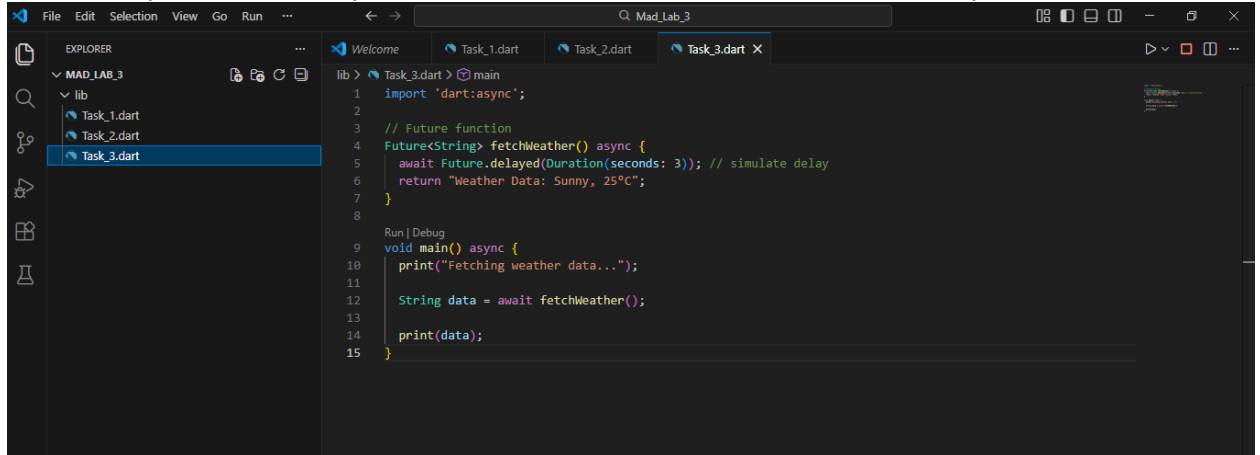
The screenshot shows a terminal window with the following output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Rifaq Ajmal\Desktop\Mad_Lab_3> cd lib
PS C:\Users\Rifaq Ajmal\Desktop\Mad_Lab_3\lib> dart run Task_2.dart
Enter number of elements: 4
Enter number 1: 4
Enter number 2: 5
Enter number 3: 3
Enter number 4: 5
Maximum number is: 5
PS C:\Users\Rifaq Ajmal\Desktop\Mad_Lab_3\lib>
```

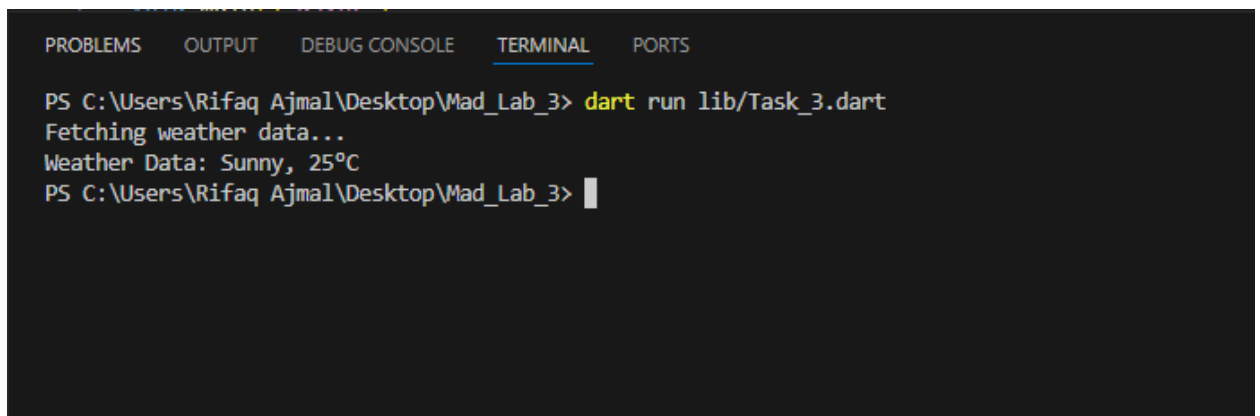
Task 3

- 3) Imagine you are developing a "Weather App" that needs to fetch data from a server. Since network requests take time, you must use a Future function to simulate this delay.



The screenshot shows an IDE with the Explorer panel on the left displaying a project named 'MAD_LAB_3' containing a 'lib' directory with three Dart files: 'Task_1.dart', 'Task_2.dart', and 'Task_3.dart'. The 'Task_3.dart' file is selected and its code is visible in the editor. The code defines an asynchronous function 'fetchWeather()' that simulates a network delay using 'Future.delayed' and returns a string representing weather data. A 'main' function calls 'fetchWeather()' and prints the result.

```
lib > Task_3.dart > main
1 import 'dart:async';
2
3 // Future function
4 Future<String> fetchWeather() async {
5   await Future.delayed(Duration(seconds: 3)); // simulate delay
6   return "Weather Data: Sunny, 25°C";
7 }
8
9 Run | Debug
10 void main() async {
11   print("Fetching weather data...");
12
13   String data = await fetchWeather();
14   print(data);
15 }
```



The screenshot shows the terminal window of the IDE. The 'TERMINAL' tab is active, displaying the command 'dart run lib/Task_3.dart' and its output. The output shows the program printing 'Fetching weather data...' followed by 'Weather Data: Sunny, 25°C' after a delay.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Rifaa Ajmal\Desktop\Mad_Lab_3> dart run lib/Task_3.dart
Fetching weather data...
Weather Data: Sunny, 25°C
PS C:\Users\Rifaa Ajmal\Desktop\Mad_Lab_3> |
```